

VR-Based Visualization of walking fatigue in simulated environment of variable PM2.5 and PM10 particulate concentration

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ABSTRACT

Urbanization and environmental pollution significantly impact human health. This impact is often imperceptible due to its long-term nature. However, virtual environments can simulate the effects of various phenomena on the human body, including poor air quality. We propose a solution that considers the reduced speed of human movement in a VR environment. It adjusts the player's movement speed according to the level of urbanization in the environment, as determined by the dispersion of PM2.5 and PM10 pollution.

Index Terms:

D.2 [Software engineering]: D.2.10 Design— [H.5]: Multimedia Information Systems —Artificial, augmented, and virtual realities; User Interfaces ; J.5 [Arts and Humanities]: Architecture—;

1 INTRODUCTION

Urbanization and environmental pollution can have a significant impact on health. One consequence of urbanization is the increased exposure to traffic and the PM2.5 and PM10 particles it produces [9]. PM2.5 and PM10 particles are known to pose significant health risks when inhaled, leading to respiratory and cardiovascular problems. These health impacts could potentially influence the speed of human movement indirectly by affecting overall physical well-being. They are also reflected in respiratory inflammation and affect oxidative stress, which is reduced in green areas [3].

Poor air quality, characterized by high levels of PM2.5 and PM10, can cause discomfort and health problems, which can reduce physical activity levels and movement speed [2,8]. Exposure to car traffic may contribute to a permanent reduction in lung function in people with asthma [5]. This suggests that virtual environments simulating poor air quality might need to account for reduced movement speeds to reflect realistic human behavior.

We propose a solution that utilizes different speeds for player movement based on the type of environment in which the player is located, specifically the degree of urbanization, by analyzing PM2.5 and PM10 dispersed molecules.

2 OUR PROPOSAL

2.1 Conceptual phase

In the initial phase of the project, meetings were held with urban planners and a pulmonologist. Based on their insights, preliminary educational goals and initial versions of the project were defined and implemented, which became the pillars for all subsequent stages of development. VR technology has been used as an educational tool, with positive perceptions [7], the use of VR is expected to improve support the ability to remember information [6].

During the development of the application, iterative testing was conducted until it was ready for demonstration. As part of the planned experiment, educational goals were defined. Visualizing the

impact on the body's capabilities is intended to raise awareness of the greater stress on the body associated with air pollution [4] compared to previous real-world experiences. The goal is to increase internal motivation to make more conscious habitats while simultaneously reducing environmental impact.

3 DATA

The input data are PM2.5 and PM10 levels. These are collected and processed according to EU standards [1]. The data comes from 1,000 locations across Europe. The readings are assigned a location, with type – 349 records for and with measurements averaged on an annual basis. The total number of measurements for the stations taken into account in 2023 is 1,554,456, an average of 4,454 for each station. A simple machine learning model was implemented. The input data is an .xml file with extracted data. A linear regression model was performed. A scalar map was constructed based on the sorted data as part of the reduction from 3D to 2D. The resultant color map was found to be a two-dimensional layered configuration (fig. 1). The map was subsequently integrated into the Unreal Engine environment. The movement of the player relative to the map depends on the environment, with reference to the concentrations of PM2.5 and PM10. Alterations in the coloration of the terrain, perceptible to the player, serve as an indication of a potential reduction in movement velocity upon entering distinct zones that are influenced by the molecules.

3.1 Implementation

The Unreal Engine, version 5.6, with the VR Expansion Plugin [1], was selected as the simulation environment. The visualization utilizes coordinates derived from a map that has been constructed previously. The distance scale has been reconfigured to ensure optimal player movement within the virtual reality environment. The implementation of a coordinate system resulted in the generation of spheres (particles) containing location data and mean PM2.5 and PM10 concentrations. The position of the spheres is determined by the coordinates, which represent specific locations. The coloration of these particles depends on the air saturation with suspended PM2.5 and PM10 particles.

For simplicity, it was decided that only extreme values and intermediate values (5 in total) would be selected for the project. This decision resulted in a reduction in the number of samples implemented. In order to enhance the clarity of the visualization, a two-dimensional hypsometric map has been implemented, with colors analogous to the PM2.5 and PM10 intensity values assigned to the spheres. The map has been implemented as a JPG file.

3.2 Interaction Method

The system operates on the HTC VIVE Focus 3 VR headset. It enables users to navigate the map at varying speeds, depending on the saturation levels of PM2.5 and PM10 at their location. The map has been graphically divided into five regions. Each color represents a different level of air pollution, and each color-coded field exerts a distinct influence on the user's rate of movement across the map. Users can directly feel the difference in movement speed depending on the dust zone in which they are. The average dust concentration

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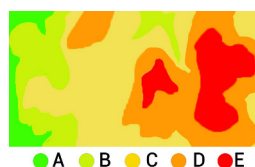


Figure 1: A part of the resulting two-dimensional map. Scale adapted to VR move. A. Good air quality, B. Average air quality, C. Unhealthy air quality, D. Very unhealthy air quality, E. Dangerous air quality.



Figure 2: Color value and its impact on the speed of movement. A - molecule colors. B - colors for the scalar map.

is also highlighted as landmarks on the map, providing reference points for the discovery of the impact of dust on the body.

As users approach designated landmarks, their speed of movement decreases. As they move away from the landmark, their speed increases until they enter the influence field of the next landmark. Players control their movements using a joystick. To visualize the duration of the influence of walking fatigue, re-acceleration is limited. It is a function of the body's exposure time, where the maximum reduction in acceleration is 0.9 of the value at the entrance to the area of influence. The total delay during the game is a maximum of 0.3 of the initial ability. This occurrence is contingent on the application of scalar weights to specific points, facilitating the average calculation of movement speeds. Note that the landmarks are located at the maxima of the scalar function. The color boundaries of the map are located at the inflection points of the function. The colors of the "particle" are subject to variation, depending on the intensity of PM2.5 and PM10 at a specific location (fig. 2). The selection of a specific particle by the user prompts the appearance of additional information regarding the type of location, along with a reading of values using the displayed interface.

The interface has been expanded with a 2D map displayed on the screen. This allows the user to know their position in the entire context of the map and to easily move between individual color fields. This is intended to allow freedom of route selection and enable the user to experience changes in speed characteristics in a manner of their choosing (fig.3).

4 EVALUATION

To evaluate the approach, interviews were conducted with groups that could use the solution for educational purposes. These researchers were actively representing:

- 1) Urban planners and architects.
- 2) Lung Disease Doctors.
- 3) Gamers who use VR technology.

A total of nine interviews were conducted. Given the limited number of cases analyzed, the IPA methodology was used to conduct in-depth interviews. Each interview consisted of three parts: the first focused on professional activities, knowledge of VR technology, and the use of programming tools. The second part of the presentation concerned the opinions of participants on the VR environment that we had designed. The third part of the study focused on opinions regarding the feasibility of implementing the solution in the workplace of the respondents' work environment.

The results of the study demonstrated that the respondents appreciated the user-friendly visual aspect of the solution and its simplicity. It was noted that some of the respondents were unable to identify a use for the proposed solution in their work. The aspects of use that were mentioned most frequently were awareness and education dur-

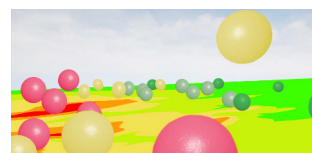


Figure 3: A map implementation

ing workshops for young people and the application of the solution to computer games. The results will be used to further develop the application.

5 FUTURE WORK

This study presents a method for testing the impact of pollution on human health. For this purpose, a VR presentation was created based on data collected from air quality measurement points. Visualized movement delays when a virtual user is exposed to suspended particles in the air. We have created a VR experience that allows users to experience a reduction in their walking efficiency depending on the degree of urbanization.

We present a method for simulating the impact of PM2.5 and PM10 molecules on human movement speed in a virtual environment. We want to establish an algorithm for walking fatigue based on medical data. In particular, it is important to incorporate knowledge from the field of lung performance and issues related to changes in emotional perception depending on the route traveled in VR. Due to the computational model, it is also important to collect data from other regions with a similar degree of urbanization.

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